

B.Tech IV Year I Semester (R20) Regular Examinations December/January 2024

GEOMETRIC DIMENSIONING AND TOLERANCES

(Mechanical Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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| (a) Define datum axis, datum target lines, datum target areas, datum target point representation. | 2M |
| (b) Define, the types of dimensioning. | 2M |
| (c) Define unit flatness, perfect flatness. | 2M |
| (d) What are the principles of dimensioning? | 2M |
| (e) Define runout and types of runouts in geometrical tolerance. | 2M |
| (f) Define any 4 categories, characteristics and symbols of geometric Characteristic Symbols. | 2M |
| (g) What is meant by filtering in surface roughness measurement? | 2M |
| (h) Define Ra, Rz, Rp, Rv, Rsk. | 2M |
| (i) Mention various types of profilometers. | 2M |
| (j) What is meant by measurement uncertainty? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) Define primary, Secondary and tertiary datum planes along with datum target areas with diagram. 5M
 (b) Explain 3 – 2 -1 principle with diagram. 5M
- OR**
- 3 (a) Elaborate degree of freedom with a example. 5M
 (b) Explain the basic principles of rule1 and rule 2 with an example. 5M
- 4 Define radial element and describe how a radial element specification is applied to a drawing. Explain with a minimum of two examples. 10M
- OR**
- 5 Define tangent plane and describe the relationship between the actual surface, the tangent plane, and the geometric tolerance. 10M
- 6 Define positional Tolerancing of Nonparallel Holes with two examples. 10M
- OR**
- 7 Define fundamental dimensioning rules applying to any of the two industrial applications parts. 10M
- 8 Explain 2D-measurement method of surface roughness (i) on cylindrical surface, (ii) on linear surface. 10M
- OR**
- 9 In the measurement of surface roughness, heights of 12 successive peaks and valleys were measured from a datum as
 Peaks- 41, 43, 40, 33, 35, 32 microns
 Valleys 32, 25, 26, 24, 19, 42 microns
 Determine the Ra Value. 10M
- 10 Differentiate conventional and CMM method of inspection with an example. 10M
- OR**
- 11 With an example, Explain in detail about how contact type profilometer works. 10M
